

[최종모델 클래스별 성능 및 혼동행렬]



Task : 1차 전처리 파일을 활용하여 학습한 모델별 성능



실행 및 진행 사항 정리

LightGbm(임계값 x)

[검증] Macro-F1=0.6854 | Weighted-F1=0.9953 | Accuracy=0.9958

※ 정상이 99%라 Accuracy는 항상 높음 → Macro-F1로 판단

[사기검출] PR-AUC=0.9685 ROC-AUC=0.9985

[클래스별 성능] (a~l=사기유형, m=정상)

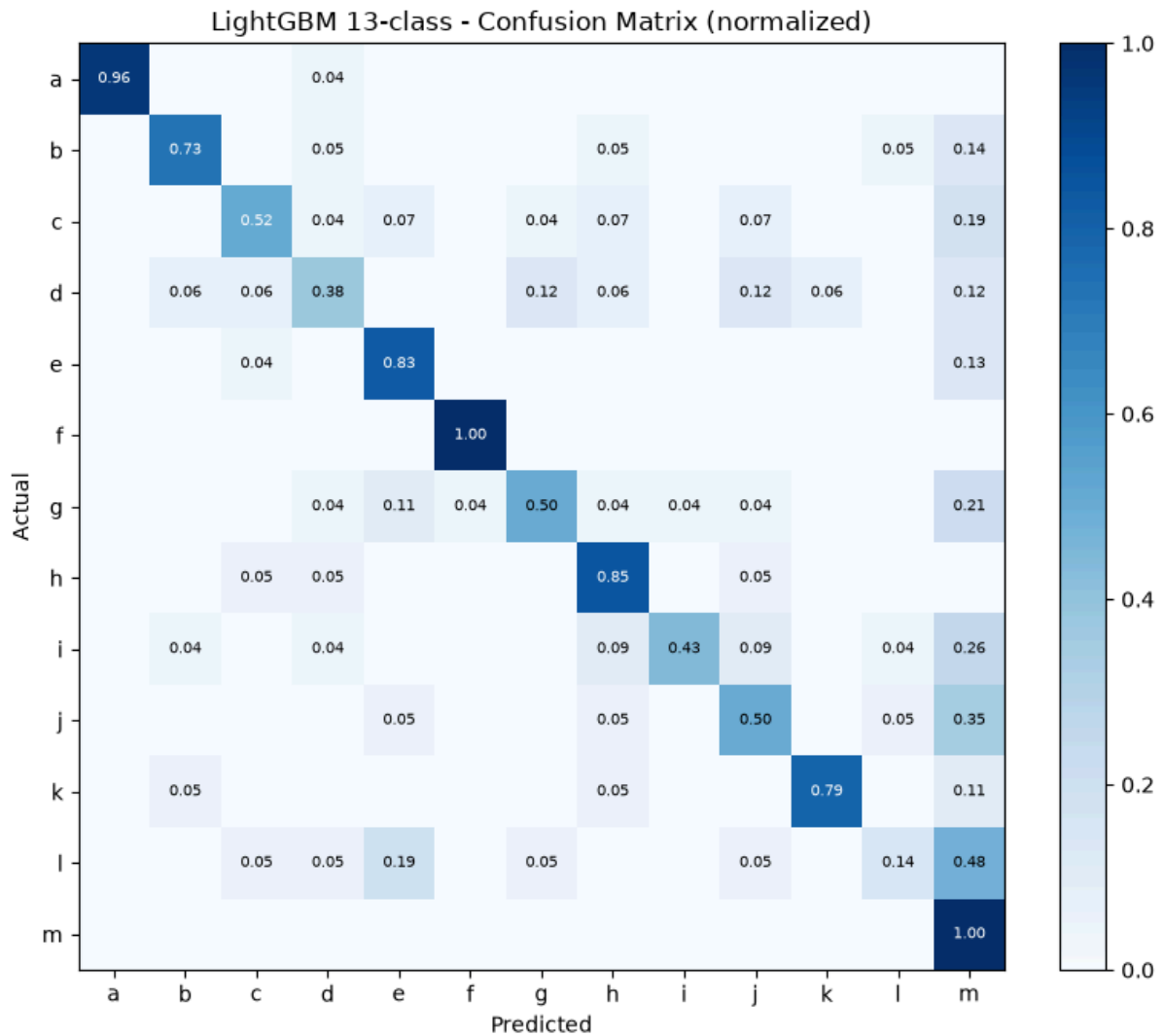
precision recall f1-score support

a	1.00	0.96	0.98	23
b	0.84	0.73	0.78	22
c	0.78	0.52	0.62	27
d	0.46	0.38	0.41	16
e	0.63	0.83	0.72	23
f	0.80	1.00	0.89	16
g	0.78	0.50	0.61	28
h	0.65	0.85	0.74	20
i	0.83	0.43	0.57	23
j	0.53	0.50	0.51	20
k	0.94	0.79	0.86	19
l	0.50	0.14	0.22	21
m	1.00	1.00	1.00	23602

accuracy 1.00 23860

macro avg 0.75 0.66 0.69 23860

weighted avg 1.00 1.00 1.00 23860



[유형별 상위 중요 피쳐 (importance top3)]

a: Time_difference_seconds, Customer_Birthyear, Account_initial_balance

b: Transaction_Amount, Transaction_Amount_abs, Customer_credit_rating

c: terminal_malicious_count, amount_to_month_max_ratio,

Transaction_Amount

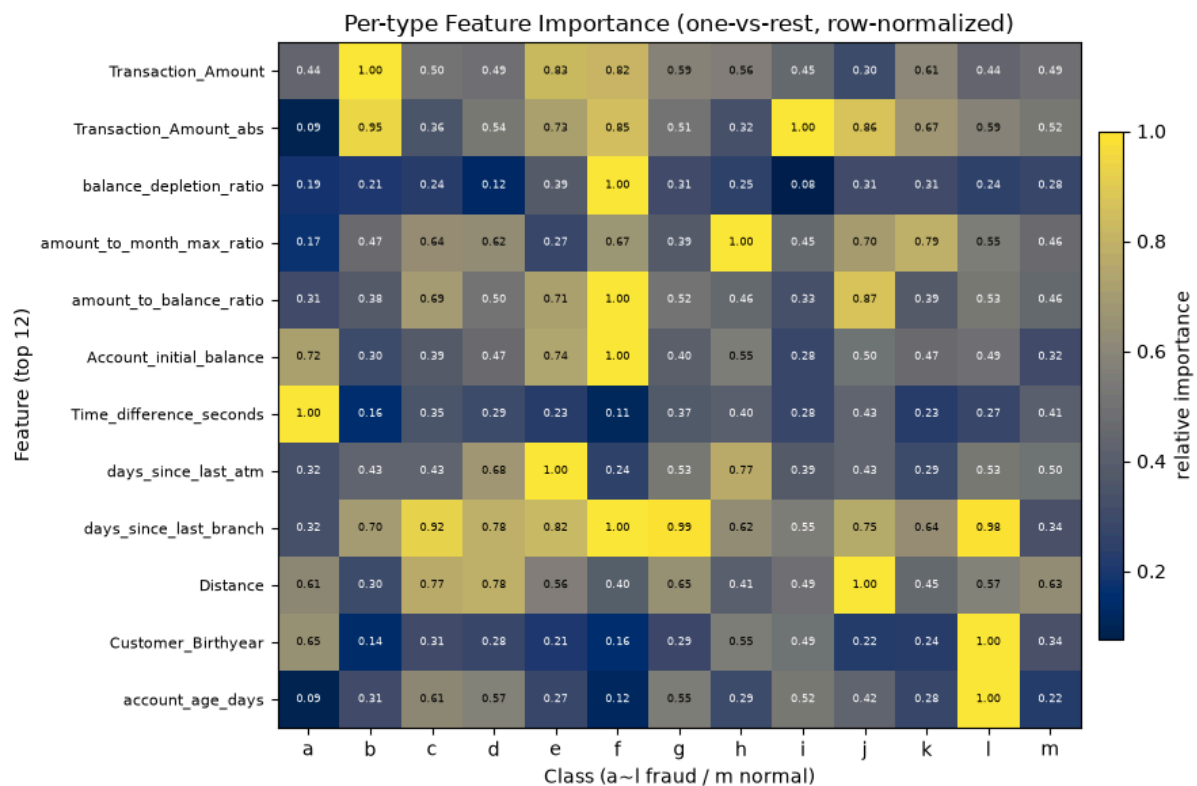
d: amount_to_month_max_ratio, Transaction_Amount, days_since_last_atm

e: Transaction_Amount, days_since_last_atm, balance_depletion_ratio

f: balance_depletion_ratio, Channel, Transaction_Amount

g: Transaction_Amount, balance_depletion_ratio, amount_to_daily_limit_ratio

h: amount_to_month_max_ratio, Account_one_month_max_amount, Transaction_Amount
i: Account_dawn_one_month_max_amount, Account_dawn_one_month_std_dev, Transaction_Amount_abs
j: Transaction_Amount_abs, Account_account_type, amount_to_balance_ratio
k: Channel, amount_to_month_max_ratio, Transaction_Amount
l: Customer_Birthyear, account_age_days, Transaction_Amount_abs
m: Transaction_Amount, Transaction_Amount_abs, Account_account_type



[유형별 SHAP] TreeExplainer 계산 (표본 2000행) ...

[유형별 상위 기여 피쳐 (SHAP top3)]

a: Time_difference_seconds, Distance, Transaction_Amount

b: Customer_credit_rating, device_network_risk_count, Channel

c: terminal_malicious_count, Channel, Transaction_num_connection_failure

d: Channel, Access_Medium, Transaction_Amount

e: days_since_last_atm, Channel, Flag_deposit_more_than_tenMillion

f: days_since_last_branch, Flag_deposit_more_than_tenMillion, days_since_resumed

g: Flag_deposit_more_than_tenMillion, Account_release_suspention, Transaction_Amount_abs

h: Account_one_month_std_dev, Account_one_month_max_amount,

amount_to_month_max_ratio

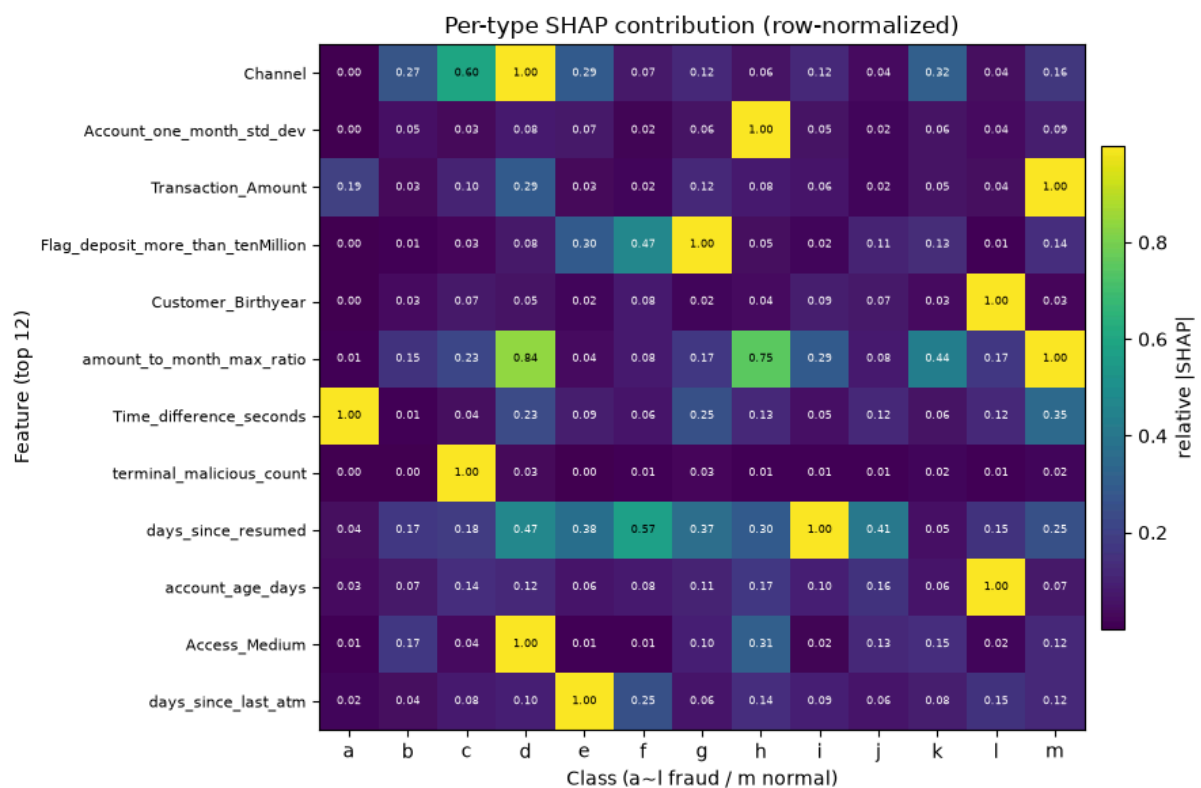
i: Account_dawn_one_month_max_amount, Account_dawn_one_month_std_dev,
days_since_resumed

j: suspended_recipient_and_withdrawal, Transaction_history_with_the_account,
Account_account_type

k: Number_of_transaction_with_the_account, Channel,
amount_to_month_max_ratio

l: Customer_Birthyear, account_age_days, Customer_loan_type

m: Transaction_Amount, Transaction_is_withdrawal,
amount_to_month_max_ratio



클래스별 성능 + 혼동행렬 해석

유형	Recall	주요 오분류처	해석
a, f	0.96, 1.00	거의 없음	여전히 최상위(EDA·SHAP 결과와 일치)
k	0.79	b로 5%	안정적
h	0.85	b·c로 소량	양호
c	0.52	m으로 19%	절반은 잘 잡지만 정상으로 새는 비중 여전

유형	Recall	주요 오분류처	해석
d	0.38	e·g·j·m으로 분산(각 6~12%)	가장 넓게 흩어짐 — 특정 유형과 헛갈리는 게 아니라 "전반적으로 확신이 없는" 패턴
g	0.50	m으로 21%	절반이 정상으로 흡수
i	0.43	m으로 26%	마찬가지로 정상 쪽 누수가 큼
j	0.50	m으로 35%	j유형 오분류의 대부분이 정상으로 감
l	0.14	m으로 48%, d로 19%	가장 취약 . 절반 가까이가 정상으로 사라짐

Two-Stage 계층분류

1단계 임계값 (기본 0.5임계값)

[1단계 임계값 스윙] 정상 vs 사기 — 낮출수록 미탐↓·오탐↑

임계값	Recall	미탐(FN)	오탐(FP)	Precision
0.05	0.9651	9	379	0.3965
0.10	0.9419	15	173	0.5841
0.15	0.9302	18	107	0.6916
0.20	0.9070	24	72	0.7647
0.25	0.9031	25	51	0.8204
0.30	0.8876	29	34	0.8707
0.35	0.8760	32	22	0.9113
0.40	0.8643	35	16	0.9331
0.45	0.8527	38	11	0.9524
0.50	0.8450	40	7	0.9689
0.55	0.8411	41	4	0.9819
0.60	0.8295	44	3	0.9862
0.65	0.8178	47	3	0.9860
0.70	0.8062	50	3	0.9858
0.75	0.7984	52	2	0.9904
0.80	0.7907	54	1	0.9951
0.85	0.7674	60	1	0.9950
0.90	0.7442	66	0	1.0000
0.95	0.7171	73	0	1.0000

[2계층 TwoStage] 1단계 정상/사기 → 2단계 유형분류 (엔진: LightGBM)

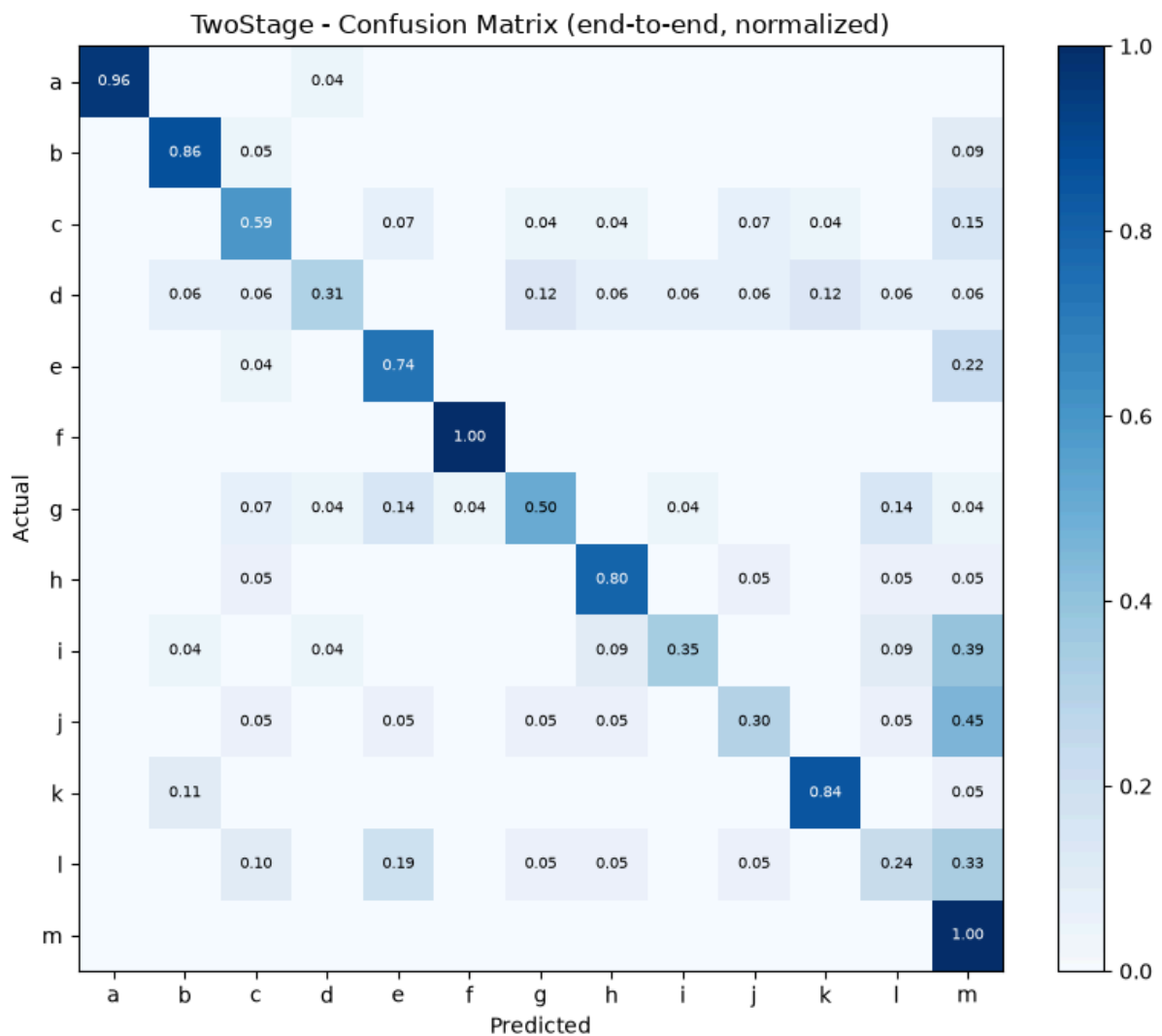
[2계층] end-to-end Macro-F1=0.6725 | 사기검출 Recall=0.8450 (임계값=0.5)

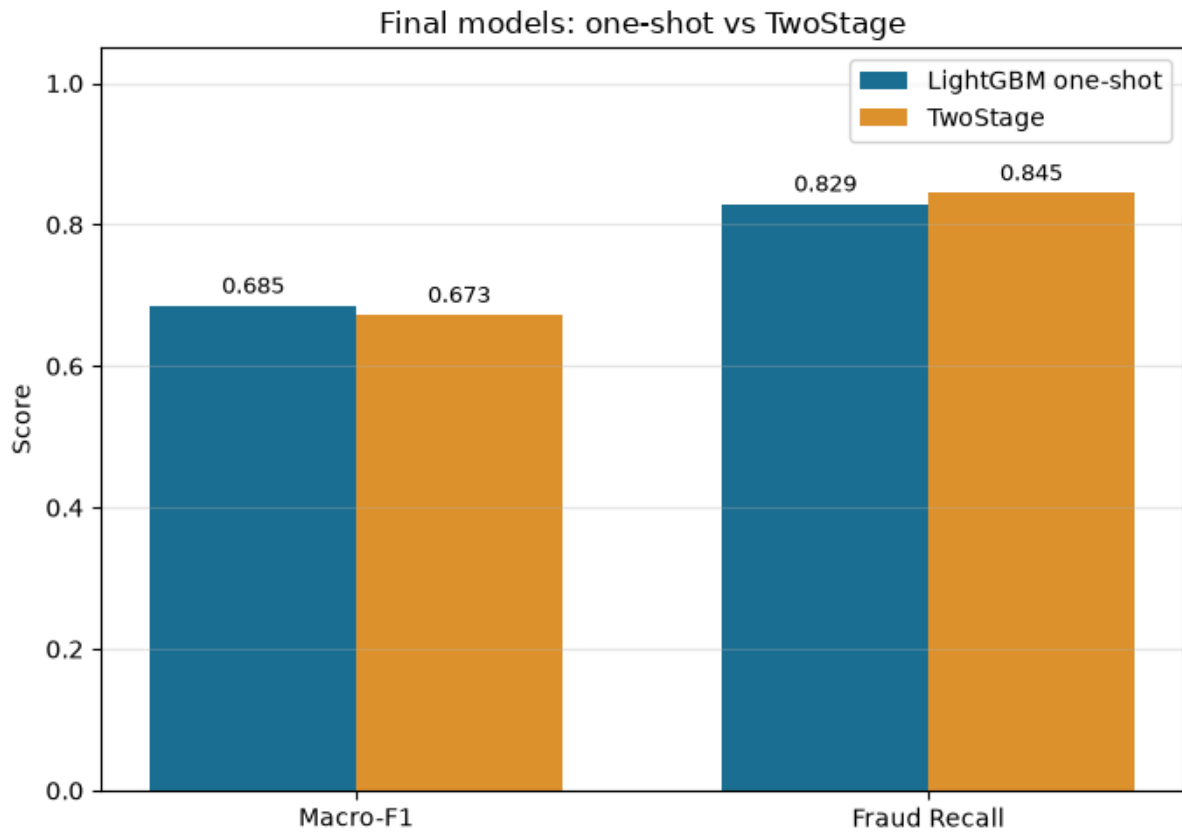
[최종 모델 비교]

· LightGBM 직접분류(한방) : Macro-F1 0.6854

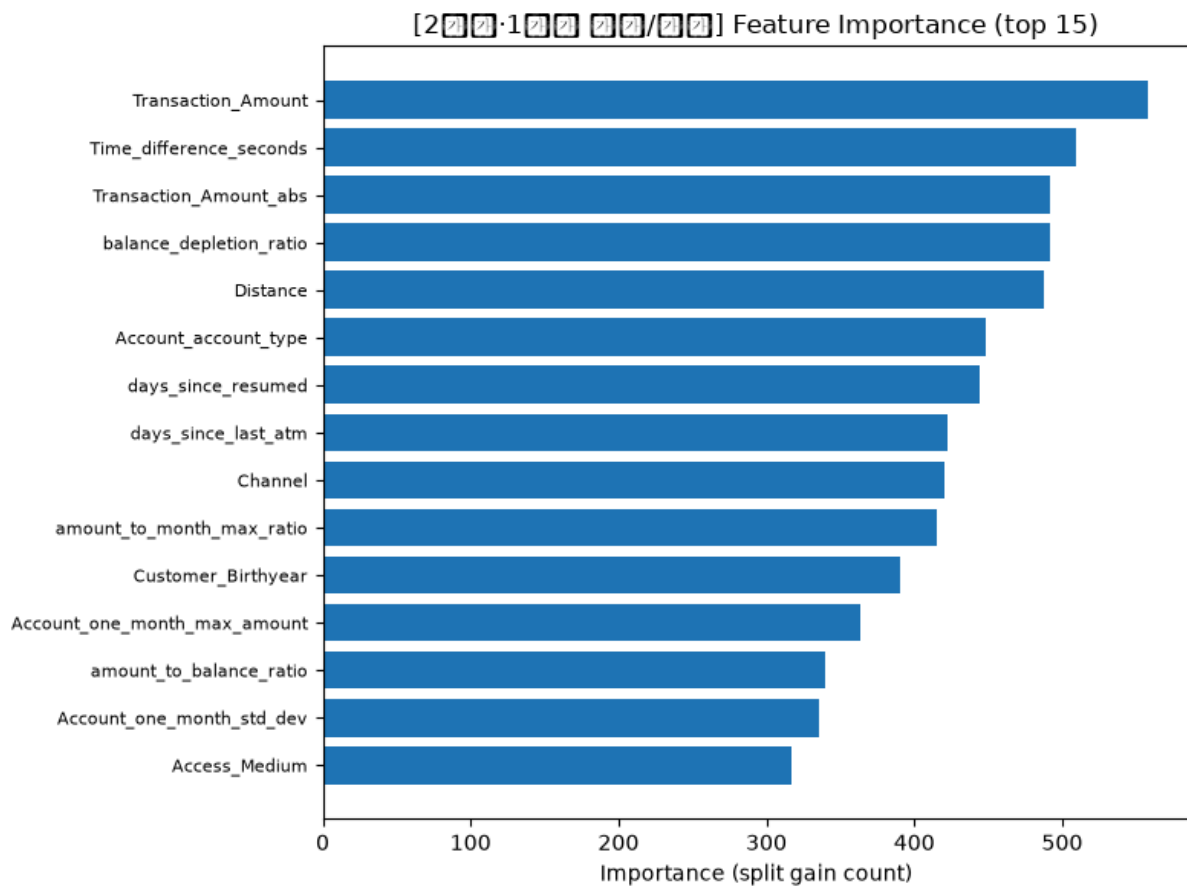
· 2계층(TwoStage) : Macro-F1 0.6725

※ 한방=성능·단순 / 2계층=임계값으로 미탐 통제·오류전파 분석 → 상황에 맞게 선택

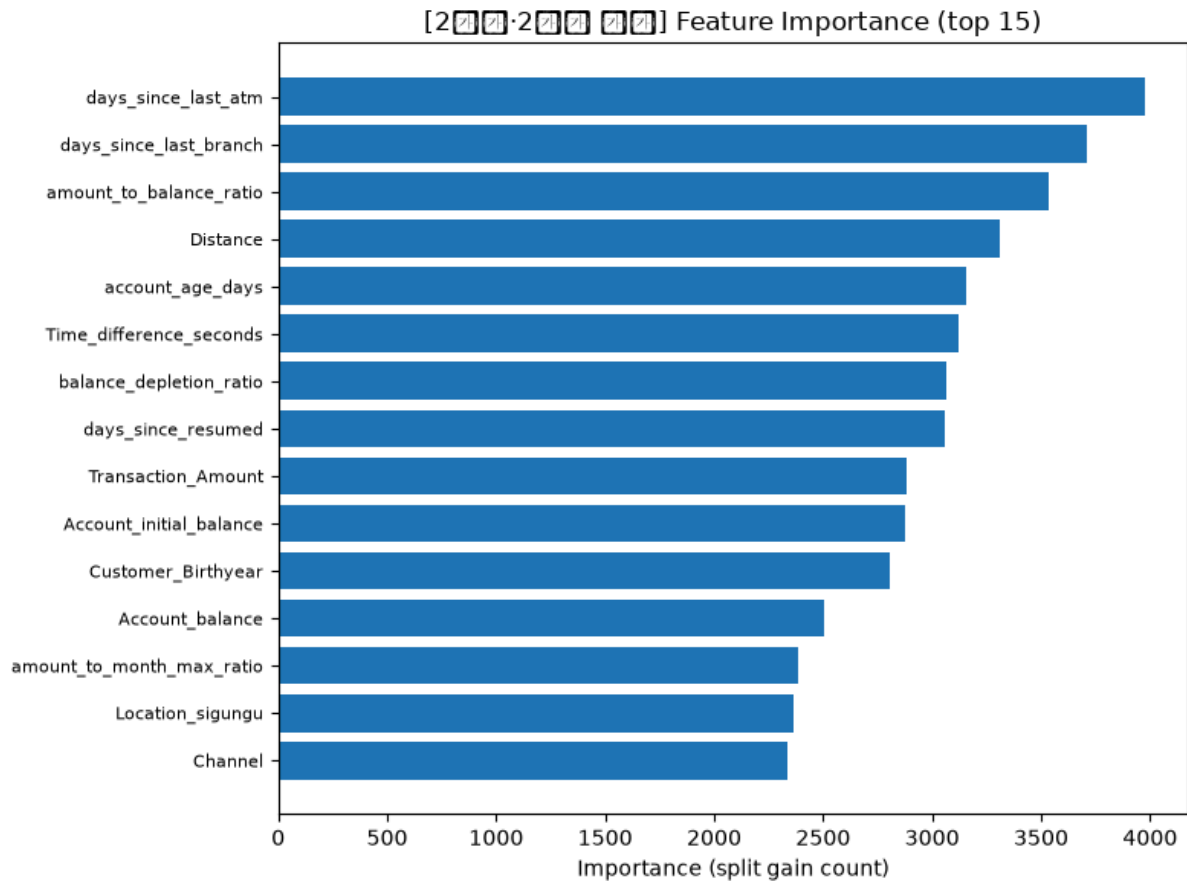




[2계층·1단계 정상/사기] [피처 중요도 top15] Transaction_Amount(558),
Time_difference_seconds(509), Transaction_Amount_abs(492),
balance_depletion_ratio(492), Distance(488), Account_account_type(448),
days_since_resumed(444), days_since_last_atm(422)

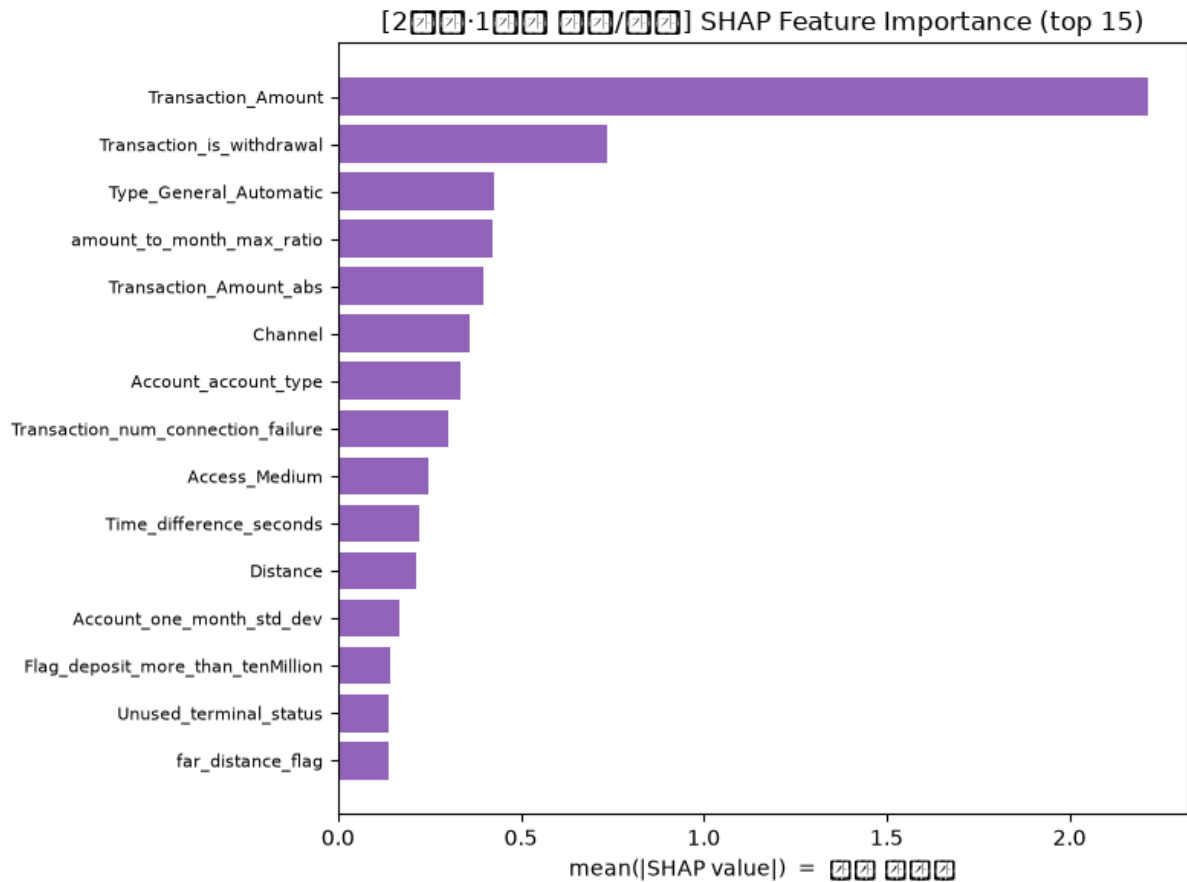


[2계층·2단계 유형] [피처 중요도 top15] days_since_last_atm(3974),
 days_since_last_branch(3710), amount_to_balance_ratio(3536), Distance(3313),
 account_age_days(3155), Time_difference_seconds(3123),
 balance_depletion_ratio(3066), days_since_resumed(3057)



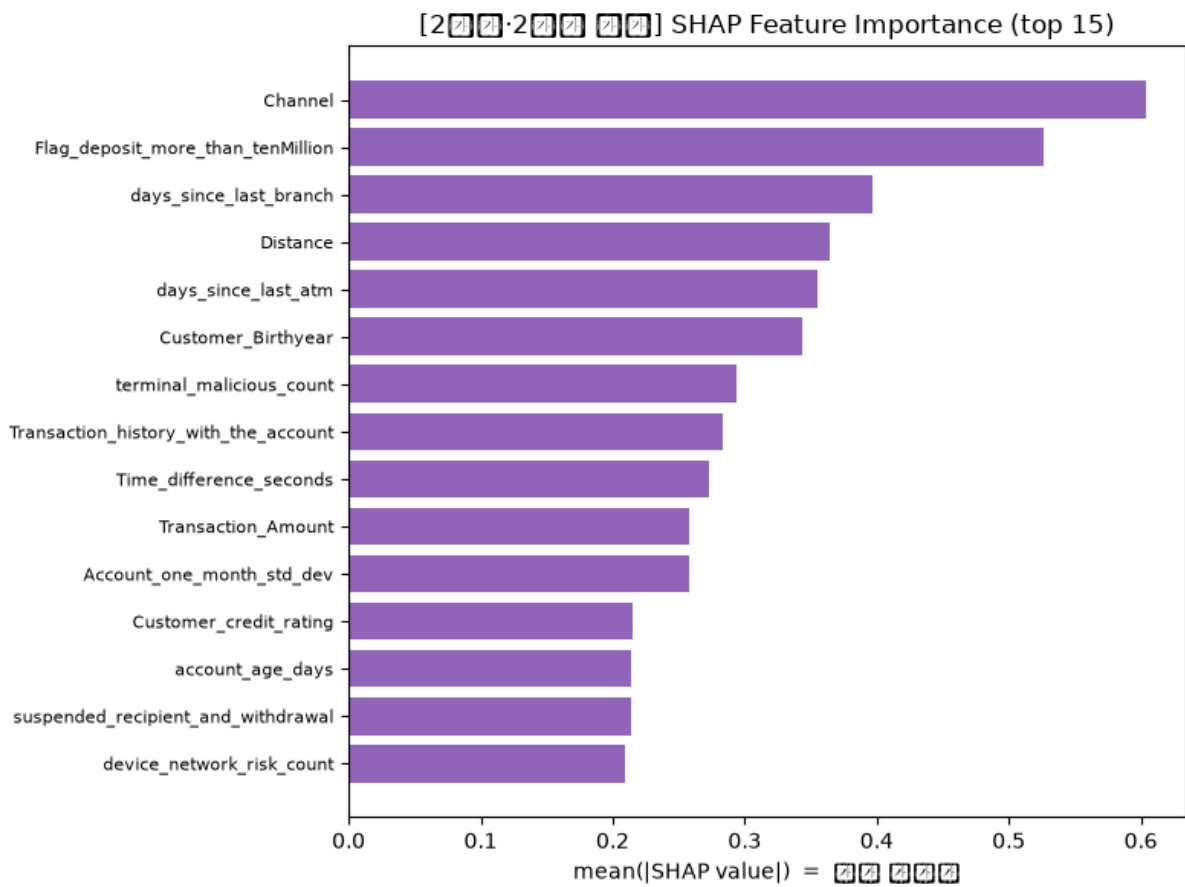
[2계층·1단계 정상/사기] [SHAP] TreeExplainer 계산 (표본 2000행) ...

[2계층·1단계 정상/사기] [SHAP top15] Transaction_Amount,
Transaction_is_withdrawal, Type_General_Automatic,
amount_to_month_max_ratio, Transaction_Amount_abs, Channel,
Account_account_type, Transaction_num_connection_failure



[2계층·2단계 유형] [SHAP] TreeExplainer 계산 (표본 258행) ...

[2계층·2단계 유형] [SHAP top15] Channel, Flag_deposit_more_than_tenMillion, days_since_last_branch, Distance, days_since_last_atm, Customer_Birthyear, terminal_malicious_count, Transaction_history_with_the_account



 **결과**